resQroute

Intelligent Hazard-Based Red Zone Identification, Carrying Capacity Assessment & Relocation Platform

SIH Problem Statement: SIH26191	**Target Stack:** React + FastAPI + PostGIS
Lead Architecture Baseline: Modular Monolith	**Core Thesis:** Shortest Route ≠ Safest Route
Primary Deliverable: Web-First MVP Platform	**Document Date:** August 2026

1. Executive Summary & Core Engineering Thesis

During natural disasters (floods, landslides, storm surges), the existence of an emergency shelter does not guarantee that vulnerable citizens can safely reach or use it. Conventional routing systems (Google Maps, Waze) optimize aggressively for shortest distance or travel time, often directing citizens directly into submerged low-lying roads or active hazard zones.

CORE ARCHITECTURAL PRINCIPLE:
*1. **SHORTEST ROUTE ≠ SAFEST ROUTE**.*
*2. **AI ASSISTS — VALIDATED DATA, SAFETY CONSTRAINTS & DETERMINISTIC ROUTING DECIDE**.*
*3. **SMS PROVIDES A FALLBACK COMMUNICATION CHANNEL — resQroute PROVIDES THE INTELLIGENCE**.*

2. System Architecture & Component Scope

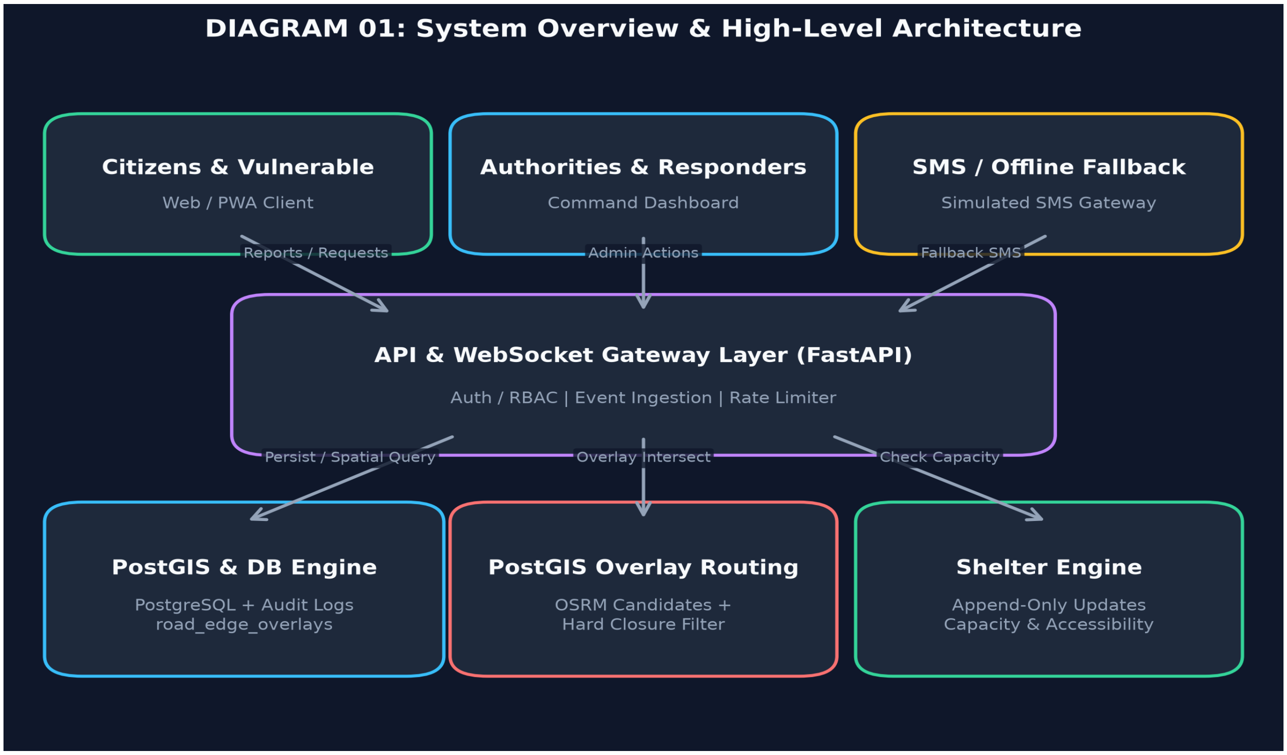

Figure 1: High-Level Modular Monolith System Architecture

Component	Classification	Technical Scope & Technology Stack
Responsive Web Map	[MVP]	React 18 + TypeScript + Leaflet / MapLibre + Tailwind CSS
API Gateway	[MVP]	Python FastAPI + Pydantic + Redis Pub/Sub WebSockets
Overlay Routing Engine	[MVP]	OSRM Candidates + PostGIS ST_Intersects Dynamic Edge Filter
Shelter Capacity	[MVP]	Append-Only shelter_updates Event Stream + Materialized View
Simulated SMS Gateway	[MVP]	Compact text response parser modal for web demonstration
Shadow ML Classifier	[ENHANCED]	NLP text report classification & image debris checking in shadow mode
BLE Mesh SOS	[ADVANCED]	Peer-to-peer experimental device mesh for zero-connectivity signals

3. Routing Decision Engine & PostGIS Dynamic Overlays

To avoid misleading terminology: OSRM's preprocessed road graph is static and does not serve as a live dynamic-risk engine by itself. Instead, `resQroute` uses OSRM to generate K-shortest candidate paths, after which PostGIS performs dynamic spatial overlay checks (`ST_Intersects`) against active closed edge buffers (`road_edge_overlays`).

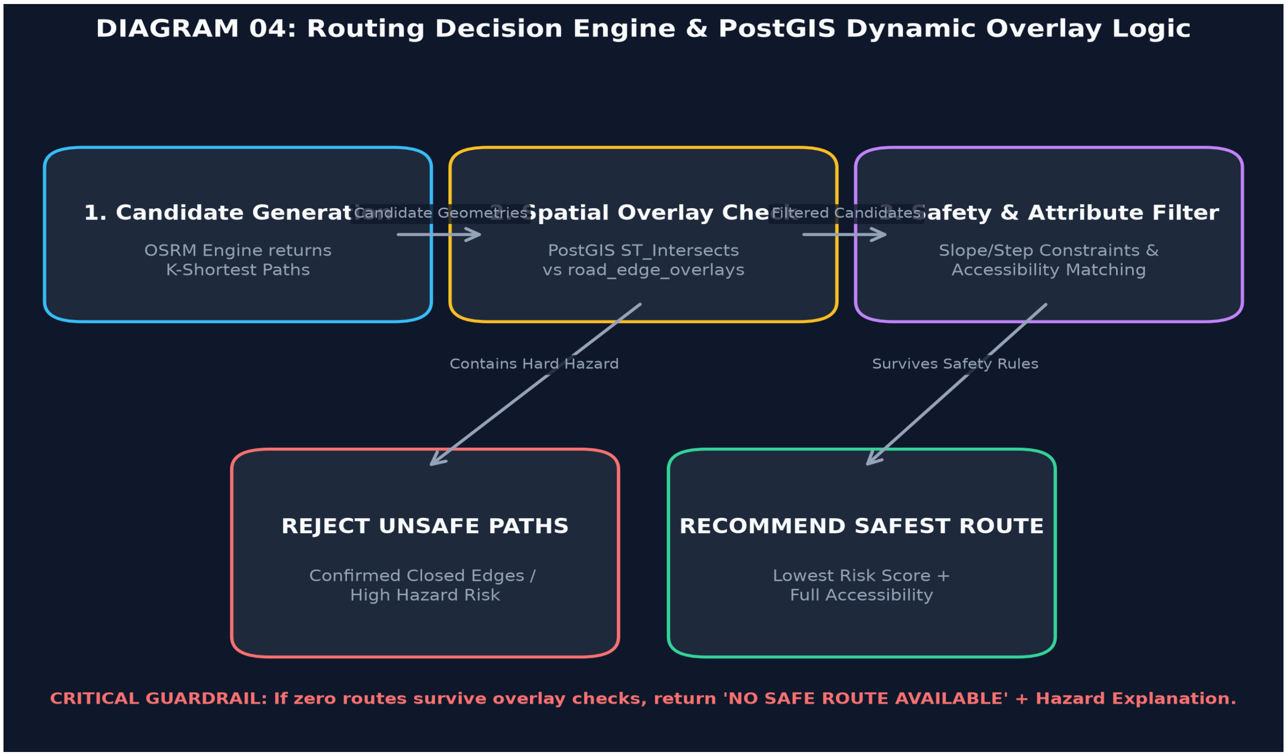

Figure 2: PostGIS Overlay Routing Decision Pipeline

4. PostGIS Data Architecture & Audit Schemas

All hazard reports, road edge closures, and shelter occupancy changes maintain strict data lineage. The data model incorporates audit tables (`road_edge_overlays`, `hazard_evidence`, `source_registry`, `audit_events`, `shelter_updates`) to ensure every published hazard identifies its source, freshness, and actor history.

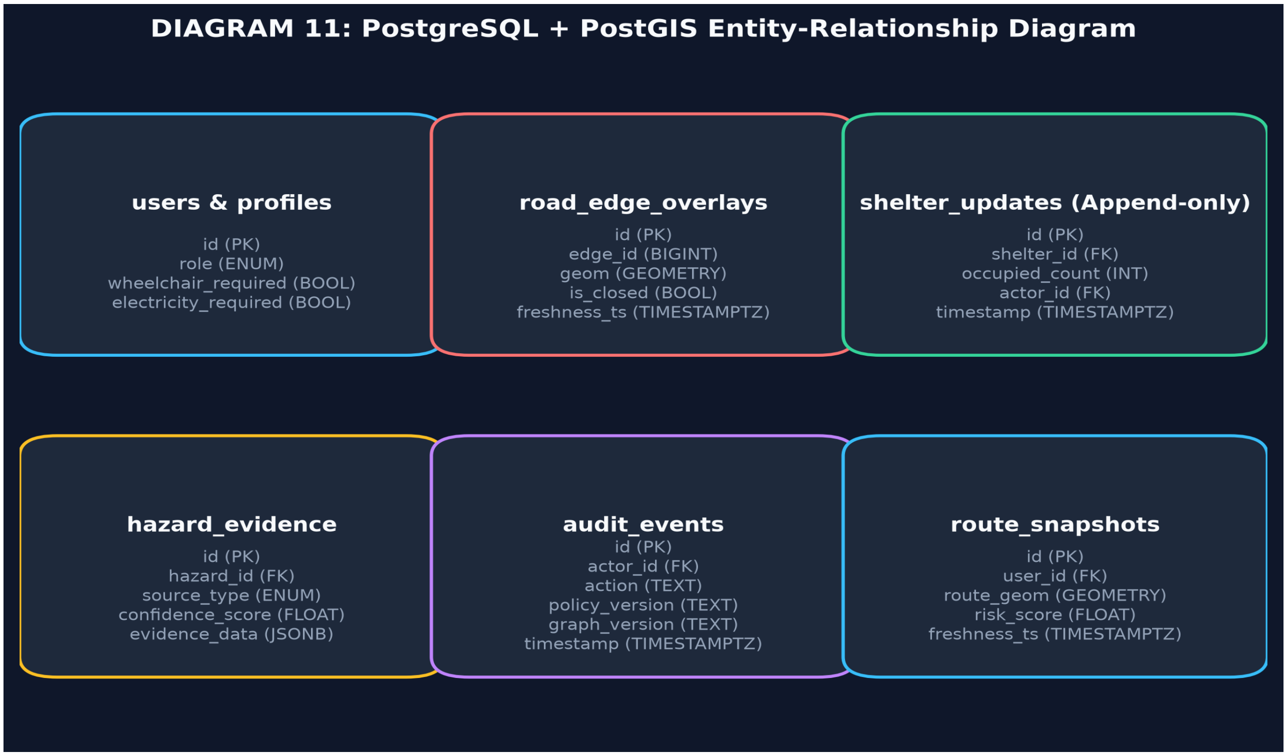

Figure 3: PostgreSQL + PostGIS Safety & Audit ER Diagram

5. Behavioral Safety Assertions & Test Suites

Suite ID	Disaster Condition	Behavioral Safety Assertion
TS-001	Road Edge #402 Closed	Confirmed closed edges MUST NEVER appear in returned evacuation routes.
TS-002	Shelter Occupancy = 100%	Full shelters MUST NEVER be recommended; auto-redirect to next nearest.
TS-003	Wheelchair Profile Active	Routes with stairs/steep slopes MUST BE REJECTED; ramp mandatory.
TS-004	Data Sync > 30m Stale	System MUST trigger explicit stale warning banner or return NO_SAFE_ROUTE.
TS-005	Unverified Citizen Report	Raw report CANNOT create hard closure without policy evidence or authority action.

6. Beginner Technical Glossary

Term	Definition
API	Application Programming Interface allowing frontend and backend services to exchange structured data.
Backend	Server-side logic and database operations processing user requests asynchronously.
Frontend	User interface layer running in web browsers built with HTML, CSS, and JavaScript/TypeScript.
GIS	Geographic Information System designed to capture, store, analyze, and manage spatial data.
GPS	Global Positioning System providing location coordinates (latitude and longitude) via satellite.
PostgreSQL	Open-source object-relational database management system offering high reliability and SQL compliance.
PostGIS	Spatial database extension for PostgreSQL adding support for geographic objects and spatial queries.
Redis	In-memory data structure store used for sub-millisecond caching and real-time WebSocket pub/sub messaging.
REST	Representational State Transfer architectural style for stateless, HTTP-based web services.
WebSocket	Full-duplex, persistent communication protocol enabling instant server-to-client broadcasts.
ML	Machine Learning algorithms that detect patterns and make predictions from historical data.
LLM	Large Language Model trained on extensive text datasets for natural language processing and extraction.
NLP	Natural Language Processing algorithms analyzing unstructured text into structured hazard records.
Computer Vision	AI models processing digital images to identify structural damage, flooding, or debris.
Routing Engine	Software algorithm calculating navigable paths between geographic points on a network graph.
Graph	Mathematical data structure comprising vertices (nodes) connected by edges (roads).
Node	A point in a network graph representing an intersection or geographic landmark.
Edge	A line segment in a network graph representing a physical road segment connecting two nodes.
GeoJSON	Standard open format for encoding geographic data structures using JSON objects.
Docker	Containerization platform packaging software code and dependencies into portable containers.
Cloud	On-demand availability of remote computer system resources and storage over the internet.

Cache	High-speed data storage layer storing transient data for rapid retrieval without database hits.
RBAC	Role-Based Access Control restricting system features based on authorized user roles.
JWT	JSON Web Token securely transmitting verified user identity claims between client and server.
BLE	Bluetooth Low Energy wireless protocol for short-range low-power device communication.
MQTT	Lightweight messaging protocol tailored for resource-constrained Internet of Things (IoT) sensors.
IoT	Network of physical devices embedded with sensors to exchange real-time environmental data.
ETL	Extract, Transform, Load data pipeline processing raw inputs into structured analytical tables.
Inference	Execution phase of a trained machine learning model generating predictions on new input data.
Training	Process of optimizing machine learning model weights on labeled historical datasets.
Validation	Evaluating machine learning model accuracy on unbiased dataset splits to prevent overfitting.
Precision	Proportion of true positive predictions among all positive predictions made by a model.
Recall	Proportion of true positive predictions identified out of all actual positive cases in dataset.
F1 Score	Harmonic mean of precision and recall measuring overall classification accuracy.
Deterministic Routing	Routing logic producing identical, predictable outputs for a given set of spatial constraints.

7. Phased Implementation Roadmap

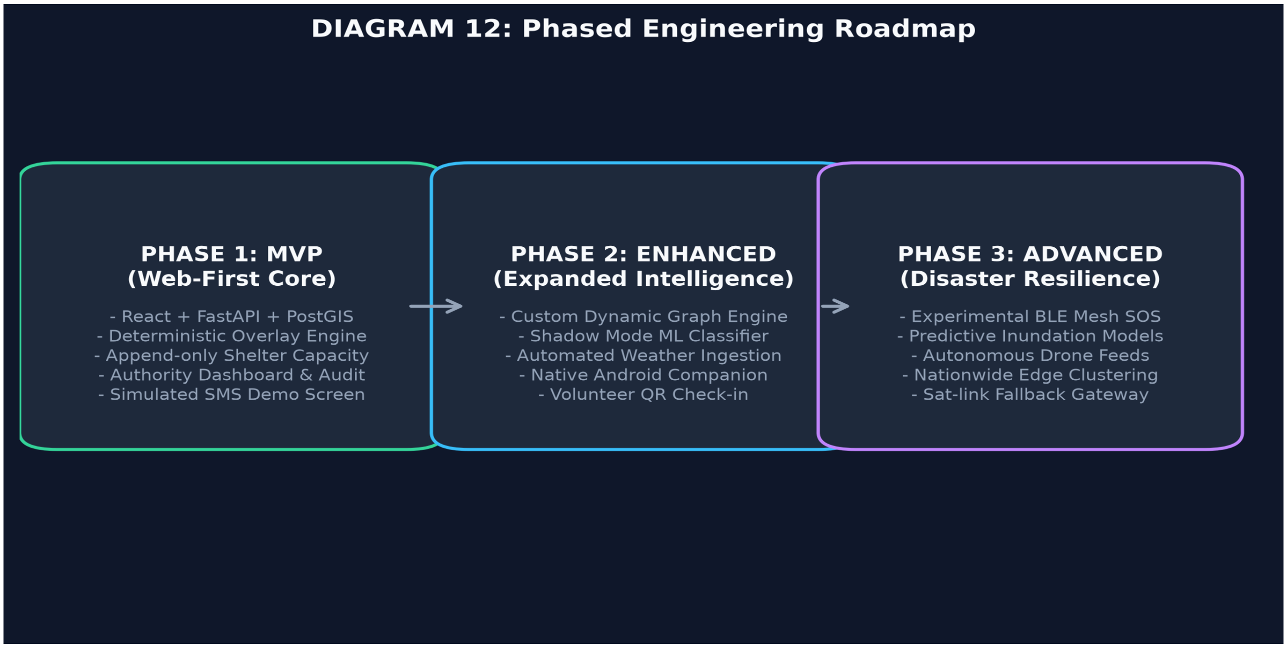

Figure 4: Phased Development Engineering Roadmap

FINAL IMPLEMENTATION PHILOSOPHY:
"Build the reliable core first. Add intelligence second. Add resilience third. Scale only after validation."

resQroute does not simply find the nearest shelter. It evaluates whether the shelter is suitable, reachable, and safe for the person who needs it.